



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION II

CANDIDATE
NAME

CENTRE
NUMBER

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INDEX
NUMBER

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CLASS

H2 BIOLOGY

9648/03

Applications Paper and Planning Question

18 September 2014

Paper 3

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/ 12
2	/ 13
3	/ 15
4 (SPA)	/ 12
5	/ 20
Total	

This Question Paper consists of **17** printed pages.

Answer **all** questions.

1. The Atlantic salmon, *Salmo salar*, is a fish in the family Salmonidae, which is found in the northern Atlantic Ocean. It has great commercial value in the food industry, and can be marketed as whole or processed. Its popularity has led to rising substitution with other non-salmon species.

A real-time Polymerase Chain Reaction (PCR) assay was developed for the identification of *Salmon salar*, allowing the detection of fraudulent or unintentional mislabelling of this species. This method can be applied to all fresh, frozen and processed products, including those that undergone intensive processes of transformation.

In this method, a pair of primers (forward and reverse primers) is designed to amplify the *ITS1* gene, a gene unique to the *Salmo* genus, during PCR. A radioactive DNA probe that would bind specifically to the *ITS1* gene is also designed.

Fig. 1.1 shows the binding positions of primers and DNA probe relative to the *ITS1* gene.

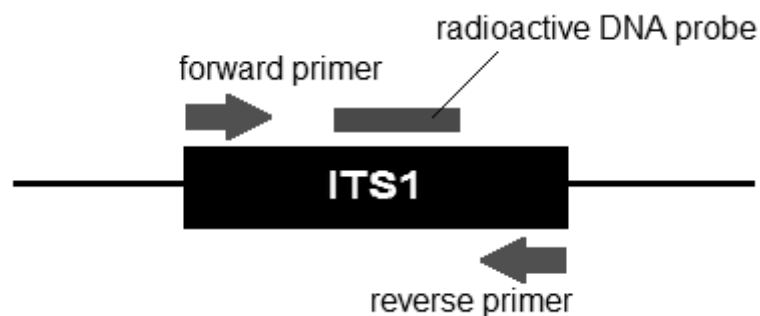


Fig. 1.1

Products labelled as 'Atlantic salmon' were bought from various markets and their DNA subjected to the PCR assay and Southern blotting.

The resulting autoradiogram is shown in **Fig. 1.2**.

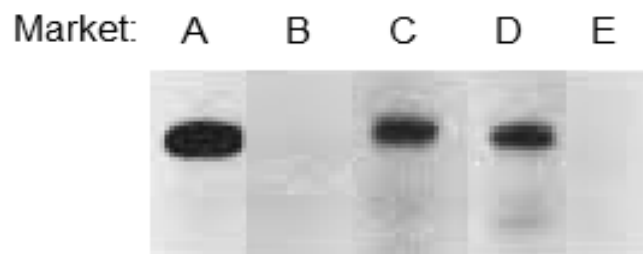


Fig. 1.2

- (a) Describe how PCR amplifies the *ITS1* gene.

[3]

1. 96°C denatures / separates the sample DNA molecules
2. Into 2 DNA strands
3. 65°C allows for the primers to anneal

4. 3' ends of *ITS1* gene
5. 72°C allows for elongation of primer / addition of dNTPs
6. To synthesis many copies of *ITS1* gene

(b) Explain why a pair of primers is needed for PCR. [2]

1. Double-stranded DNA is anti-parallel
2. Taq polymerase can only synthesise DNA in 5' to 3' direction
3. Each primer binds to 1 strand of DNA
4. allowing for both strands to serve as template for replication

(c) Identify and explain which market(s) does not sell fish from the *Salmo* genus. [3]

1. Markets B and E
2. No bands in autoradiogram
3. Absence of *ITS1* gene in DNA
4. Primers did not bind to DNA fragment
5. DNA not amplified during PCR

Fig.1.3 shows the amount of *ITSR* gene during PCR.

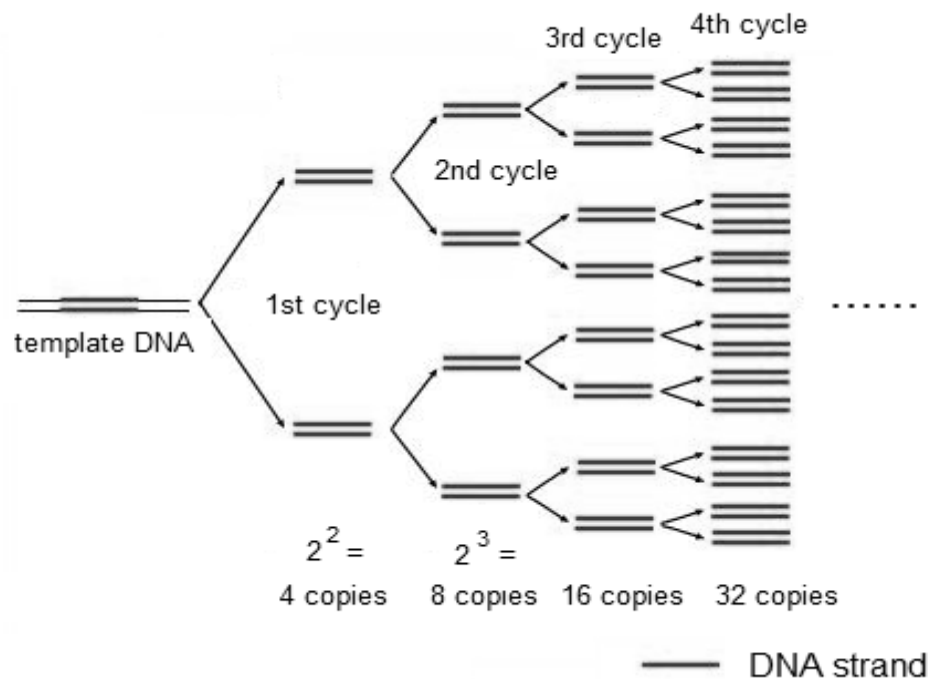


Fig. 1.3

- (d) Indicate the number of *ITS1* genes present at the end of 18 cycles. Show **all** your working clearly. [1]

$$2^{19} / 2 = 262144$$

262144 *ITS1* genes

- (e) Discuss the limitations of using PCR for the authentication of *Salmo* genus. [2]

1. Too sensitive, contaminant DNA may be erroneously amplified to give a wrong conclusion
2. Lack of proofreading function of Taq polymerase, base misincorporation may lead to wrong conclusion

- (f) Describe how the product synthesised by PCR is similar to that by DNA replication in eukaryotes. [1]

1. Both products consist of 1 parental strand and 1 newly synthesised strand
2. Both products are DNA molecules

[Total: 12]

2. Adenosine deaminase (ADA) deficiency is an autosomal recessive metabolic disorder that causes immunodeficiency. Affected individuals are unable to produce the enzyme ADA. Without this enzyme, T lymphocytes cell count is compromised, and the immune system cannot defend against pathogens.

Gene therapy, first conceptualised in 1972, is a possible course of treatment for ADA deficient individuals. Although early clinical failures led to many dismissing gene therapy, recent clinical successes have bolstered new hope in this approach.

Fig. 2.1 shows the processes involved in the gene therapy treatment of ADA deficiency.

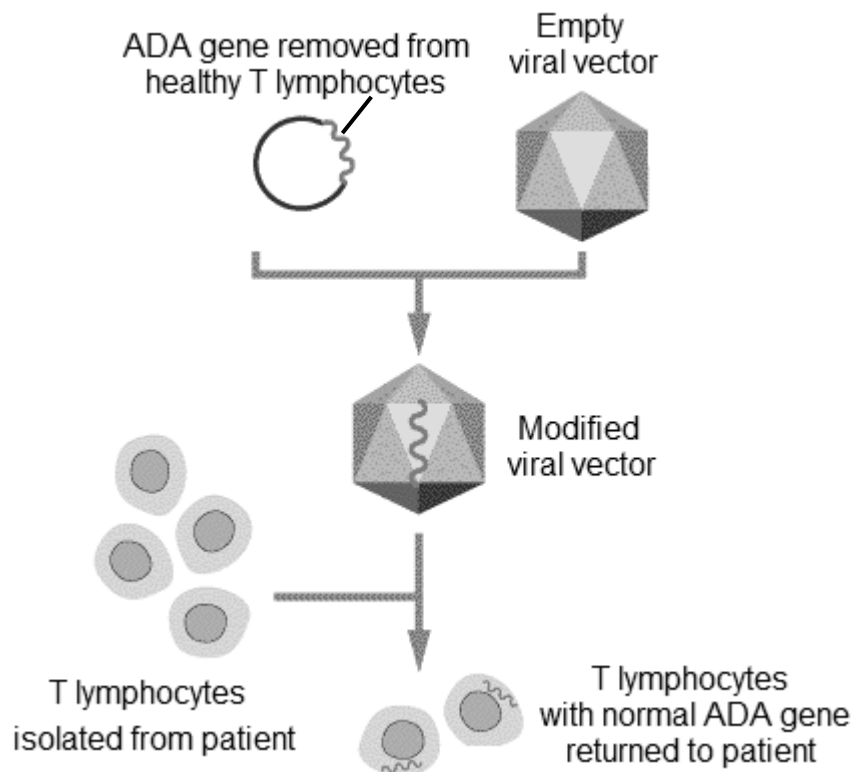


Image adapted from Klug & Cummings (1997)

Fig. 2.1

- (a) Suggest why the early clinical trials of this treatment approach failed. [2]

1. Viral vectors are destroyed by the immune system
2. Viral vectors carry viral genome, cause disease in subject
3. Not all T lymphocytes took up the therapeutic allele, insufficient protein synthesised to give therapeutic effect
4. Therapeutic allele not integrated into chromosome, resulting in sporadic expression / digested by cytoplasmic enzymes

- (b) Explain the benefits of using viral vector in gene therapy. [2]

1. Can target delivery of therapeutic allele to desired cells

2. Therapeutic allele can be integrated into the chromosome, allowing for long term stability
3. Therapeutic allele can be integrated into the chromosome, allowing for transfer to daughter cells during cell division
4. Therapeutic allele can be integrated into the chromosome, do not require multiple applications

(c) Explain why individuals treated by this method of gene therapy would still have ADA-deficient children. [2]

1. Only somatic cells are infected with modified viral vector
2. Gametes do not contain normal ADA allele
3. Can pass on recessive allele to offspring

Nine ADA deficient patients were administered with different doses of the modified viral vectors and empty viral vectors. Efficacy of the gene transfer was determined six days after the administration of respective vector. The results are shown in **Table 2.1** below.

Table 2.1

Patient	Gender	Age / years	Dose / pfu	Presence of ADA mRNA	
				Treated with empty viral vector	Treated with modified viral vector
1	F	25	2×10^7	No	No
2	M	44	2×10^7	No	No
3	M	23	2×10^8	No	No
4	M	21	2×10^8	No	No
5	F	34	2×10^9	No	Yes
6	M	40	2×10^9	No	No
7	F	19	2×10^9	No	Yes
8	F	33	2×10^{10}	No	Yes
9	M	22	2×10^{10}	No	No

(d) Describe the main trends shown in **Table 2.1**. [2]

1. No transcription of ADA gene in Patient 1 to 9 upon treatment with empty viral vector
2. Transcription of ADA gene only observed for high dose of viral

vector above 2×10^9 pfu

3. At high dose of viral vector, transcription of ADA gene was only observed females

- (e) Suggest why patients are also treated with empty viral vectors. [1]

Serves as a control / basis of comparison for treatment with modified viral vector

Shows that empty viral vector do not express ADA gene

Another approach for the treatment of ADA deficiency involves bone marrow transplant from a genetically-matched donor. This, unlike the approach shown in **Fig. 2.1**, can provide a permanent cure with a single treatment.

- (f) (i) Suggest why bone marrow transplant is obtained from a genetically-matched donor. [1]

Decrease tissue rejection

Decrease immune response

- (ii) Explain why a single bone marrow transplant can provide a permanent cure for ADA deficiency. [3]

1. Bone marrow contains blood stem cells

2. with normal ADA allele

3. Blood stem cells can divide indefinitely

4. and differentiate into T lymphocytes under appropriate conditions

[Total: 13]

3. Stem borers are insects that feed on the leaves and reside in the stem of plants, causing damage to many crops such as maize. An insecticidal toxin (Bt toxin) extracted from the bacterium, *Bacillus thuringiensis*, is found to be able to kill the stem borers but is harmless to other animals, including humans. The Bt toxin was used as an active ingredient in pesticides spray commonly used by farmers. However, there is limited success in completely killing all the stem borers on the plant.

In recent years, maize plant has been genetically modified to express a gene for Bt toxin. Tiny beads of tungsten were coated with plasmids containing the Bt gene and fired at very high speed at the maize cells to form Bt maize.

- (a) Suggest why pesticides containing Bt toxin cannot effectively eliminate the stem borers from the plant. [1]

stem borers residing in the stem are not exposed to pesticides sprayed on the plant

- (b) Explain how Bt gene is inserted into the plasmid. [4]

1. use same restriction enzyme
2. cut plasmid DNA
3. cut at restriction sites flanking the Bt gene
4. produces sticky ends
5. anneal via hydrogen bonds
6. ref. to complementary base pairing
7. DNA ligase
8. catalyse formation of phosphoester bond between Bt gene and plasmid

- (c) Explain the benefit of growing Bt maize. [1]

1. resistance to stem borers, results in increase crop yield
2. eliminate the need to spray pesticides, hence save cost

A study was carried out to see which types of Bt maize and their protein products would be most efficient against three species of stem borer. The stem borers were allowed to feed on nine types of maize (**A–I**), modified with Bt genes. The graph below shows the leaf areas damaged by the stem borers after feeding on maize leaves for five days.

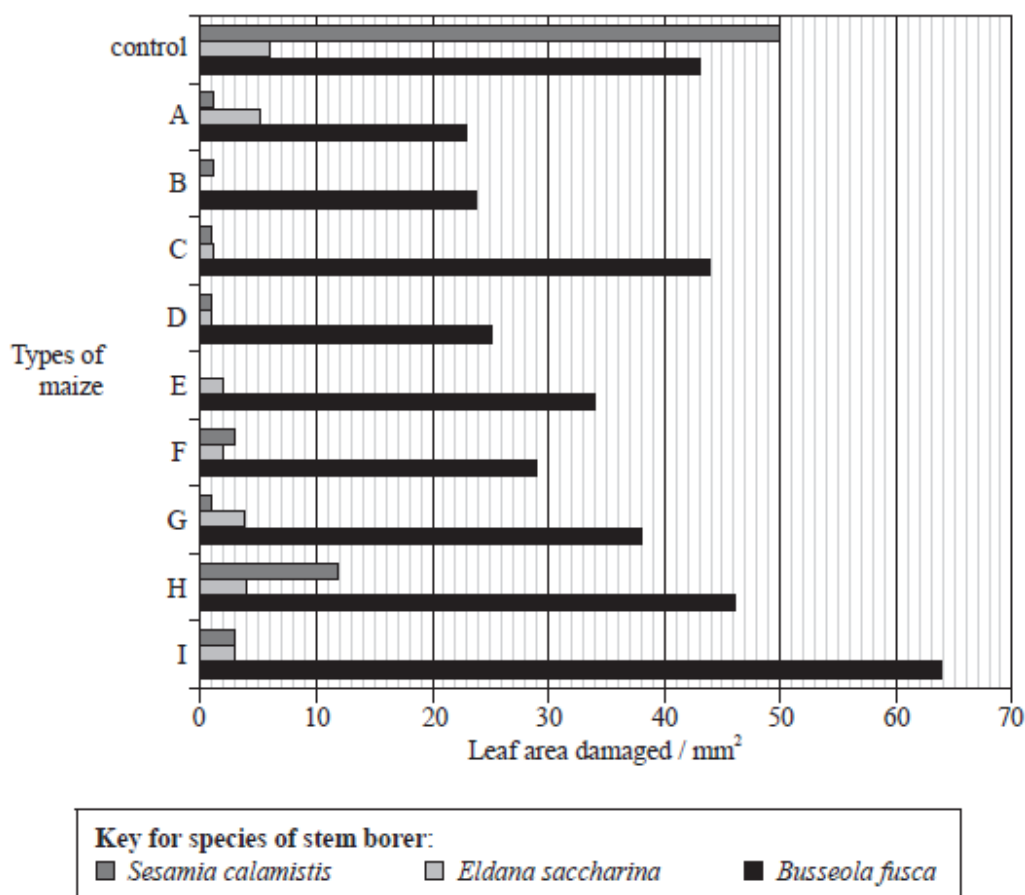


Image adapted from Mugo et al., 2005

Fig. 3.1

(d) With reference to Fig. 3.1,

(i) identify which species of stem borer was most successfully controlled by the Bt maize;

[1]

Sesamia calamistis

(ii) explain your answer in (d)(i).

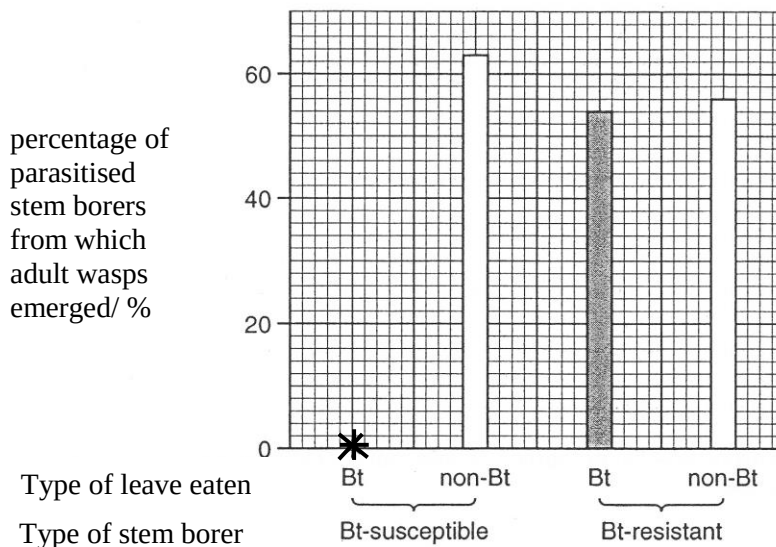
[2]

1. caused most leaf damage in control plants
2. 50 mm² leaf area damaged
3. a significant decrease in leaf damage in all types of Bt maize
4. 1-13 mm² leaf area damaged

Some stem borers are resistant to Bt toxin. The Bt-resistant stem borers were used to find out the effect of Bt maize on the food chains. The stem borers may be parasitized by a wasp, which lays its egg inside them. The developing wasp larvae feed on the tissues of the stem borers and eventually emerge as adult wasps, killing the stem borers.

Groups of Bt-susceptible and Bt-resistant stem borers that were parasitised by wasps were fed either leaves of a Bt maize or leaves of a non-Bt maize. The percentage of parasitised stem borers from which the adult wasps emerged was then recorded.

The results of the investigation are shown in **Fig. 3.2**.



key

* the stem borers died before the wasps completed their development and emerged

Image adapted from University of Cambridge Local Examinations, 2011

Fig.3.2

(d) With reference to **Fig. 3.2**,

(i) distinguish the effect of growing Bt maize and non-Bt maize on the wasp population. [3]

For Bt-susceptible stem borers,

- 1. Growing Bt maize will decrease wasp population**
- 2. 63% emergence from non-Bt maize but 0% emergence from Bt maize**

For Bt-resistant stem borers,

- 3. Growing Bt maize will have no effect/ slight decrease on wasp population**
- 4. 54% emergence from Bt maize and 56% emergence from non-Bt maize**

(ii) describe how the results prove that Bt toxin kills wasp larvae. [1]

- 1. 2% less emergence from Bt resistant stem borers that feed on Bt maize**

(e) Suggest the ecological implications of growing Bt maize. [2]

1. affects organisms that depend on stem borers such as wasps, thus disrupt the food chain
2. potentially change the Bt maize to become invasive/ super weeds
3. Bt toxin selects for stem borer resistant to Bt, results in changes in the gene pool
4. Cross contamination of genes can potentially introduce the Bt gene to other native species

[Total: 15]

4. Planning Question

Cellular respiration is a process in which cells produce the energy they need for survival. In the presence of oxygen, most organisms undergo cellular respiration to break down sugars and store the energy in the form of adenosine triphosphate (ATP).

You are required to plan, but **not** carry out, an investigation into the effect of the type of carbohydrates on rate of respiration in yeast.

Fig. 4.1 shows a datalogger with carbon dioxide sensor probe that can be used to measure the concentration of carbon dioxide produced.

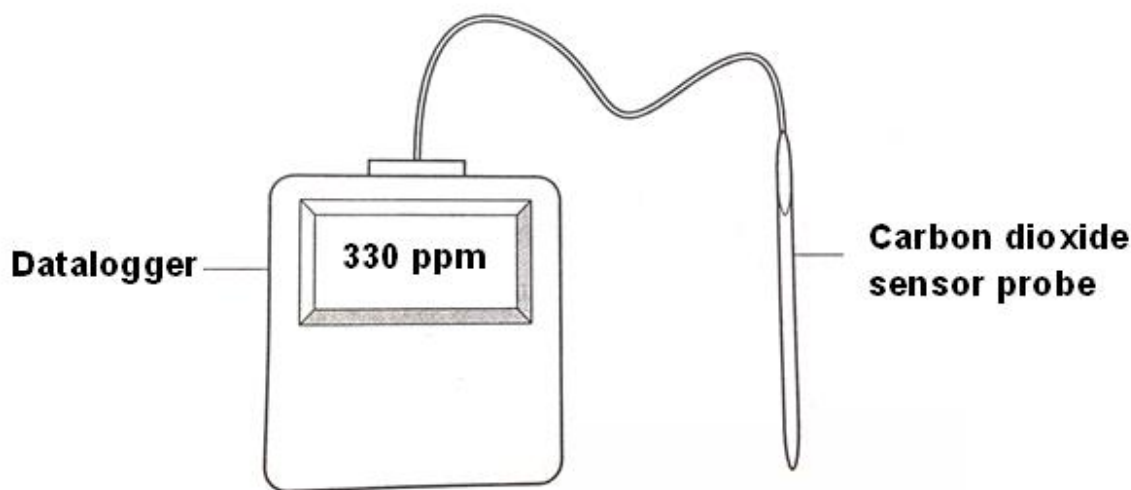


Fig. 4.1

Your planning must be based on the assumption that you have been provided with the following equipment and materials, which you must use:

- yeast suspension,
- 1% starch suspension,
- 1% sucrose solution,
- 10% glucose solution,
- rubber bung with an opening for probe
- Datalogger with carbon dioxide sensor probe
- timer, e.g. stopwatch,
- thermometer,
- a variety of different sized beakers, measuring cylinders or syringes for measuring volumes.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

A. Definition of Question

i. Aim

To investigate the effect of the type of carbohydrates on rate of respiration in yeast

ii. Relevant background knowledge of reaction/ process

1. Aerobic respiration: $\text{Carbohydrate} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
2. Oxygen is used as the final electron acceptor during oxidative phosphorylation
3. Combines with proton and electron to form water
4. Carbon dioxide is released as the by-product of oxidative decarboxylation
5. Glucose is main respiratory substrate that is used during glycolysis
6. Sucrose and starch have to undergo hydrolysis to form glucose

iii. Relate method of measurement with dependent variable

7. Datalogger with carbon dioxide sensor probe can measure concentration of carbon dioxide produced
8. in a fixed period of time
9. is directly proportional rate of respiration

iv. Describe and explain expected trend

10. Glucose gives highest rate of respiration/ Sucrose & Starch lower rate of respiration
11. used directly for glycolysis
12. Sucrose is disaccharide
13. Starch is polysaccharide
14. need to break glycosidic bond to give glucose molecules

B. Experimental Procedure and Data Manipulation

B1. Experimental variables

- Independent variable : 19. Type of substrate: Glucose, Sucrose, Starch
- Dependent variable : 20. Concentration of carbon dioxide produced in 10 minutes
- Controlled variables :

	Controlled Variable	Appropriate Quantity
21.	volume of carbohydrate used	5 cm ³
22.	concentration of carbohydrate used	1 %
23.	Volume of yeast	2 cm ³
24.	temperature	37°C / optimum temperature
25.	Duration of experiment	10 min

B2. Experimental Procedures

i. Pilot experiment

Conduct a pilot experiment to determine suitability of apparatus, optimum conditions and amount of materials used.

ii. Experimental Procedure

A: annotation on labelled diagram

26. Preparation of 1% glucose solution, i.e. Add 9cm³ of distilled water to 1cm³ of 10% glucose solution

27. Add 2 cm³ of yeast suspension and

28. 5 cm³ of glucose solution in the test tube

29. Seal test tube with rubber bung

30. Insert the carbon dioxide sensor probe through the rubber bung into the test tube

31. Place the test tube with yeast suspension and glucose solution to 37 °C in water bath

32. Measure the concentration of carbon dioxide produce over 10 minutes

33. Record the readings in appropriate table

34. Repeat experiment for sucrose solution and starch suspension

iii. Accuracy and reliability of results

35. For each type of carbohydrate, repeat the procedure two more times/ to obtain three replicates

36. Repeat the entire experiment two more times

iv. Negative Control

37. A negative control is subjected to the same factors (i.e. 2 cm³ yeast suspension) as that for the experiment, except the carbohydrate is replaced with equivolume of distilled water

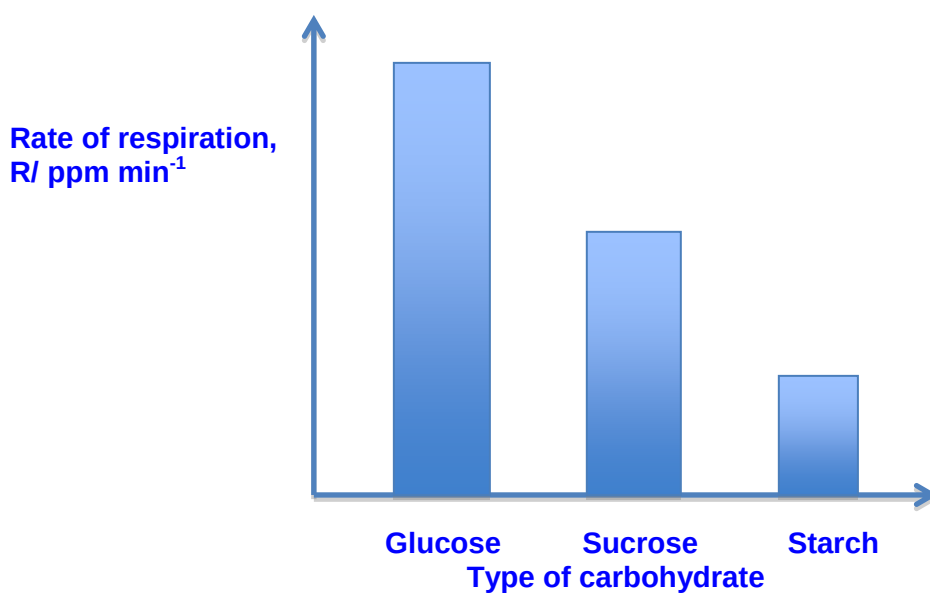
38. It is expected that there is no carbon dioxide produced regardless of incubation time. This proves that carbohydrate is indeed used as the respiratory substrate

v. Data manipulation

Table showing the effect of type of carbohydrate on rate of respiration

Type of carbohydrate	39. Concentration of carbon dioxide produced, c/ ppm;				40. Rate of respiration, R/ ppm min ⁻¹ ;
	c ₁	c ₂	c ₃	— c	
Glucose					
Sucrose					
Starch					

Graph showing the effect of type of carbohydrate on rate of respiration



41. Bar graph: glucose-highest rate, Sucrose & starch- lower rate

vi. Safety precautions

42. Use a piece of rag to hold the beaker containing hot water (when preparing water bath) to prevent burns / scalds

43. Yeast suspension can be an irritant. Wear goggles to prevent yeast suspension from getting into your eyes and wear gloves to avoid contact with yeasts

44. Dry hands before switching on the datalogger to prevent electrocution.

Free Response Question

Write your answers on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

5. (a) Micropropagation refers to the use of tissue culture techniques to propagate plants. Describe how this is carried out. [9]

1. Cells from meristematic regions of a plant are excised
2. and sterilised before transferring to culture vessels
3. containing essential minerals / vitamins / amino acids / fixed carbon (any 2)
4. Cells grown in culture to form callus tissue
5. Number of calli is increased via subculturing
6. Callus can be induced to differentiate by varying the ratio of plant growth regulators
7. Low level of cytokinin and high level of auxin triggers root growth
8. High level of cytokinin and low level of auxin triggers shoot growth
9. Organised callus tissues form a plantlet
10. which is acclimatised in enriched soil
11. before transferring to a green house to grow into a whole plant

- (b) *“The Human Genome Project (HGP) was one of the great feats of exploration in history... (giving) the complete genetic blueprint for building a human being” – Human Genome Research Institute.*

Discuss the main objectives of this project. [4]

1. To map all human genetic markers by identification of their chromosomal location
2. To construct a detailed physical map of the entire human genome
3. To determine the base sequence of all 24 human chromosomes
4. To develop a technology for the management of human genomic information
5. To serve as an umbrella for similar mapping and sequencing of the genomes of other organisms

- (c) Using a named example, explain how Restriction Fragment Length Polymorphism (RFLP) facilitates disease detection.

[7]

1. Identify carrier / heterozygotes of
2. sickle-cell anemia
3. RFLP can be used to distinguish normal β -globin allele from the mutant β -globin allele
4. The mutant allele arises due to a base-pair substitution
5. That destroys one of the (*DdeI*) restriction site on the β -globin gene
6. This prevents the restriction enzyme from cutting that particular restriction site
7. Following Southern blotting / binding of radioactive DNA probe
8. Normal β -globin allele would have 3 restriction fragments
9. Mutant β -globin allele would have 2 restriction fragments

[Total: 20]